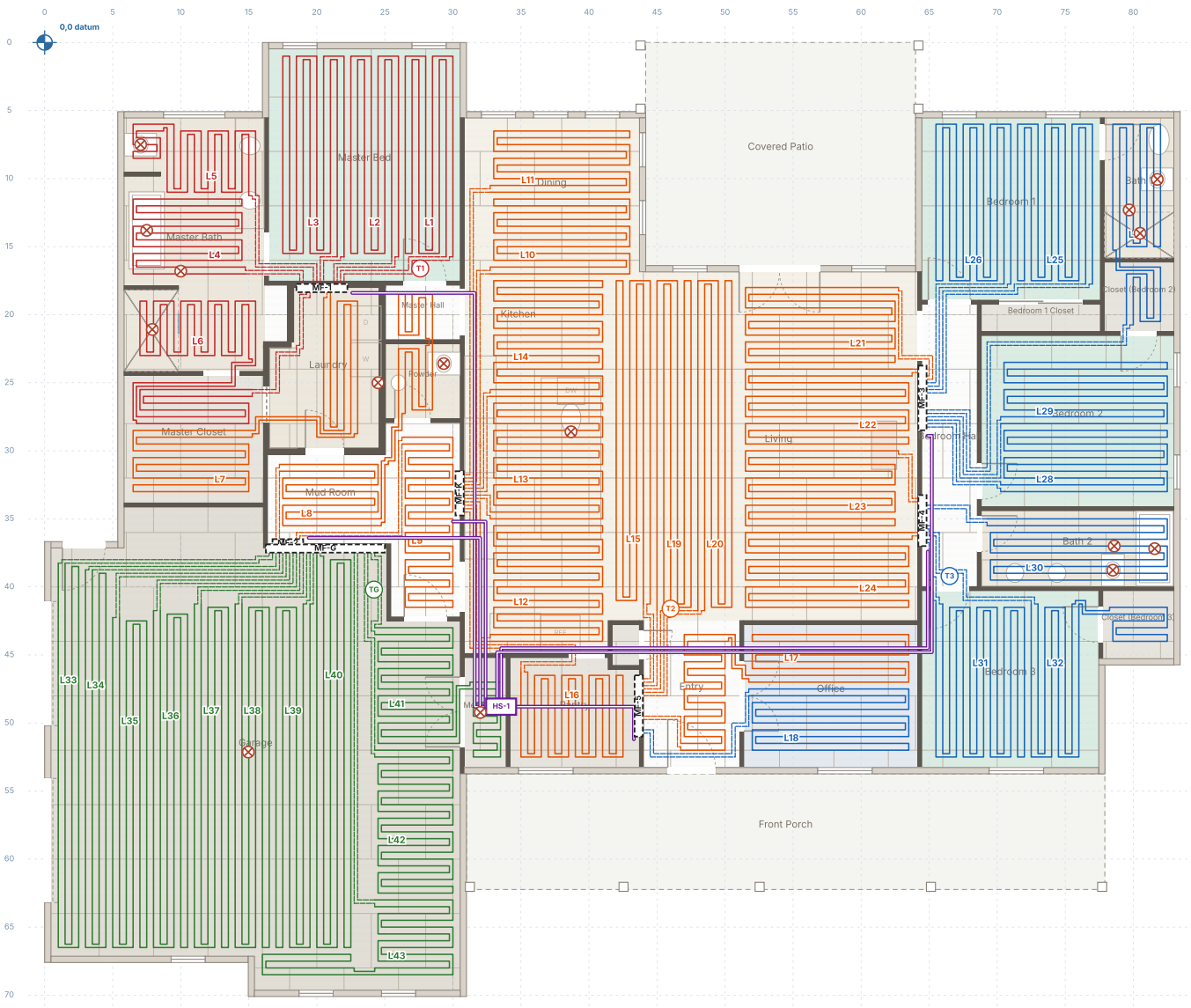
REAR · PATIO




FRONT · PORCH

Zone 1, Master suite (T1) · Zone 2, Great room and utility (T2) · Zone 3, Bedroom wing and office (T3) · Zone G, Garage (held at 50°F) (TG)

----- leaders, supply and return side by side 3" apart · 1" feed pair from HS-1, under the boards · manifold cabinet over its row of 3" ports

4' x 4' nub boards in a running bond from the datum; dots are the 3" puck grid

43 loops, 7,109' of 1/2" PEX, 7 manifolds, no loop over 200'. · Red zone 1, orange zone 2, blue zone 3, green garage. · 54 kBtu/h at design, 110°F supply. · EPS25 Super 4 nub boards on the 3" grid from the datum; nothing crosses; tube under every fixture.

draft

# R-1 Radiant · 64 Stone Sheep Circle

PEX loops on 3" nub boards under the slab: 200' loops, nothing crossing, seven manifolds, four thermostat zones, flows at Powell, WY design (-11°F). App version v23. From the owner's plans (Larson House, 6/11/26).

## Notes

1. The layout is drawn on the boards' own grid. EPS25 Super 4 nub boards (4' x 4', 5" thick: 4" of EPS, R-18, under a 1" grid of pucks on 3" centres) go down in a running bond with their puck rows on the 3" lines from the datum at 0,0 (the garage-door wall's outside face and the master bed's rear wall line, as on every sheet), so every tube on this sheet sits in a channel between pucks. Lay the first board's corner at the datum and the rest follow; a board cut at a wall keeps its grid.
2. Tube: 43 loops of 1/2" O2-barrier PEX, none over 200' with its leaders and the 2' rises at the cabinet (L6 is the longest at 191', L5 the shortest at 118'). Fields at 6" o.c. (every other channel), supply and return side by side so each loop's two ends leave its field together; leaders run as 3" pairs through doorways and along walls to their manifold. Nothing crosses anything: the whole layout was checked tube by tube on the grid. One continuous coil per loop, no fittings in the slab; tag both ends with the loop number at the cabinet. The loops on one manifold are planned to like lengths so they balance on their flow meters (MF-1 118'-191', MF-2 144'-165', MF-G 147'-187', MF-K 159'-187', MF-5 156'-189', MF-3 152'-191', MF-4 153'-180'); MF-1's spread is the master bath, whose three small fields are boxed in by the toilet and shower walls and can neither join each other nor share rows with the bedroom.
3. Under a wall only where drawn, straight across and only for the length drawn: under the master bed wall into MF-1, under the kitchen wall into MF-K, under the entry wall into MF-5, under the living room wall into MF-3 and MF-4, under the master hall wall into the powder room, and under the bath 1 wall from the bedroom 2 closet. Mark every wall crossing on the boards before the pour so the plate anchors and shots stay 12" off it; everywhere else the leaders go through the doorway.
4. The tube runs under everything that sits on the floor: the kitchen base cabinets, island, fridge and range, the washer and dryer, the tubs, showers, vanities and toilets, the hearth. It lies about 5" below the top

## Open items

- L32 (Bedroom 3, Closet (Bedroom 3)): the floor carries all but about 345 Btu/h of the room at -11°F with 110°F water. Add a small electric towel warmer, or run the boiler hotter on the coldest days.
- Not tubed: the coat closet (about 6 sq ft), the bedroom 1 bypass closet (about 17 sq ft) and the 3' x 10' notch at the garage's north wall beside the master closet (about 31 sq ft). Each is too small for a loop of its own at the lengths on its manifold, and none can be added to a neighbouring loop without crossing another; all three sit inside heated floor and borrow from it.
- Boiler model: HS-1 is a condensing boiler on natural gas or propane (decided 2026-10-13). Pick a combi or a boiler with an indirect tank if it should also make the domestic hot water and replace P-1's 50 gal heater; propane needs the boiler's LP conversion kit.
- Garage: kept at 50°F on its own manifold, pump and thermostat. Decide whether it runs a glycol mix behind a plate heat exchanger, which protects it if the house loses power in a cold snap with the doors open.
- Flooring: the loops assume hard-surface floors. Carpet in a bedroom needs about 10°F warmer water for that loop.
- Cabinet depths on S-1: MF-3 and MF-4 recess into the hall's 2×4 west wall (3-1/2" deep; fur locally if the chosen manifold needs more), MF-K into the kitchen wall and MF-5 into the entry wall.
- E-1: an outlet at each of the seven manifold cabinets for the actuators (MF-1, MF-2, MF-G, MF-K, MF-5, MF-3, MF-4), and the boiler, pump and zone controller on circuit 27.
- The EPS25 Super 4 nub boards come 5 to a bundle (80 sq ft), 3 bundles minimum: 55 bundles cover the 4,167 sq ft slab with 5% for cuts. Confirm the lead time; the board layout can start at any corner as long as the grid holds the datum.

of the slab, so nothing fastened to the floor reaches it; only the walls and their plates, the posts and the manifold port rows keep it out. Where P-1 brings a drain, a tub trap, a closet flange or a vent stack up through the slab (15 places, each marked with a crossed circle) shift the one run it meets over a channel for the length of the fitting, so the tube stays 3" clear of the flange or drain body; the showers' floors are heated like the rest.

5. Manifolds, seven stations, each with a 3" port position per tube end in one row along its wall (the schedule gives every loop's two port numbers, counted from the low end of the row): MF-1 (laundry, on the north wall west of the dryer, 7 loops) serves Master Bed, Master Bath, Master Closet, Laundry; MF-2 (mud room, on the south wall (garage wall), west end, 2 loops) serves Master Hall, Powder, Mud Room; MF-G (garage, on the mud room wall (two 7-loop manifolds side by side), 11 loops) serves Mechanical, Garage; MF-K (mud room, east wall (kitchen wall), south of the powder room, 6 loops) serves Kitchen, Pantry; MF-5 (pantry, on the east wall (entry wall), 5 loops) serves Kitchen, Living, Entry, Office; MF-3 (bedroom hall, west wall north of the fireplace, 7 loops) serves Living, Bedroom 1, Bath 1, Closet (Bedroom 2), Bedroom 2; MF-4 (bedroom hall, west wall south of the fireplace, 5 loops) serves Living, Bath 2, Bedroom 3, Closet (Bedroom 3). HS-1 in the mechanical closet feeds all seven on six 1" insulated pairs under the boards, 192' each way in all; MF-2 and MF-G share one pair rising in the garage wall.
6. The leaders rise straight up out of the slab at their port positions into the cabinet (sleeve each rise in 12" of pipe insulation where it leaves the concrete) and bundle up to the manifold; they may be gathered sideways in the last 12" below the cabinet. Cabinets are recessed in the kitchen and entry walls (MF-K, MF-5) and the hall wall (MF-3, MF-4) and surface-mounted on the laundry's sound wall (MF-1) and both faces of the garage wall (MF-2, MF-G).
7. Zoning is by thermostat: four zones (master suite, great room and utility, bedroom wing and office, garage) on 24 V actuators at each loop's manifold port, so a room can get its own thermostat later by rewiring, without new tube.
8. Flow: 11.2 gpm with every loop calling, each loop set on its flow meter at a 10°F drop. Design heat 53,627 Btu/h, of which 11,066 is the garage at 50°F; 7% is added for heat lost down through the R-18 boards and out the slab edge. Rooms with no field of their own are heated by the leader bundles running through them: Dining (228' of

leaders for 3,771 Btu/h), Bedroom Hall (115' of leaders for 422 Btu/h), Bedroom 1 Closet (2' of leaders for 86 Btu/h). Nothing heats the Coat Closet (30 Btu/h); they borrow from the rooms around them.

9. Water: 110°F supply at -11°F outside, reset lower as it warms, so a condensing boiler stays condensing. Floor surfaces stay at or under 100°F. At that supply L32 (Bedroom 3, Closet (Bedroom 3)) falls about 345 Btu/h short at design; see the open items.
10. Pump: the hardest loop is L27 at 2.0 ft in the tube; add about 3 ft for the feed pair, manifold and actuator. A constant-ΔP ECM circulator at about 5 ft covers the house; the garage has its own.
11. Under the boards: 10-mil vapor barrier over the compacted base, lapped and taped; R-10 (2") rigid foam at the slab edge. The 4" slab goes over the pucks with the tube tops about 1" below the surface of the boards; no mesh ties are needed, the pucks hold the tube. Pressure test every loop at 100 psi water or 60 psi air for 24 hours before the pour and keep it under pressure through the pour. Photograph every room with a tape in frame before concrete.
12. HS-1 is circuit 27 on E-1 (200 A panel in the garage); E-1 also gives each manifold cabinet an outlet for its actuators and the zone controller.
13. P-1's water lines, vents and drains are all in the base course below the boards and the feed pairs (cold PEX 1'-0" down, hot 1'-1 1/2", drains below both), so the boards go down whole, with no channels cut for plumbing.

## Loop schedule

Ports are the supply and return positions counted from the low end of the manifold's row. Leader is each way. Heat includes the share lost downward. Pick a row to find the loop.

| Loop | Zone | Rooms | Manifold | Ports | In field | Leader | Total | Btu/h | Flow | Head | Floor |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L1 | 1 | Master Bed | MF-1 | 14/13 | 121' | 12' | 147' | 964 | 0.20 gpm | 0.3 ft | 76°F |
| L2 | 1 | Master Bed | MF-1 | 12/11 | 121' | 9' | 140' | 917 | 0.20 gpm | 0.3 ft | 76°F |
| L3 | 1 | Master Bed | MF-1 | 8/7 | 152' | 6' | 164' | 1,082 | 0.22 gpm | 0.4 ft | 76°F |
| L4 | 1 | Master Bath | MF-1 | 1/2 | 104' | 8' | 120' | 1,385 | 0.28 gpm | 0.5 ft | 81°F |
| L5 | 1 | Master Bath, Master Bath | MF-1 | 6/5 | 90' | 14' | 118' | 1,344 | 0.27 gpm | 0.5 ft | 81°F |
| L6 | 1 | Master Closet, Master Bath | MF-1 | 3/4 | 163' | 14' | 191' | 1,930 | 0.39 gpm | 1.4 ft | 81°F |
| L7 | 2 | Laundry, Master Closet | MF-1 | 9/10 | 172' | 3' | 178' | 1,481 | 0.30 gpm | 0.8 ft | 78°F |

| Loop | Zone | Rooms | Manifold | Ports | In field | Leader | Total | Btu/h | Flow | Head | Floor |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L8 | 2 | Mud Room, Powder, Master Hall | MF-2 | 1/2 | 149' | 7' | 165' | 816 | 0.20 gpm | 0.4 ft | 75°F |
| L9 | 2 | Mud Room | MF-2 | 7/8 | 110' | 17' | 144' | 721 | 0.20 gpm | 0.3 ft | 75°F |
| L10 | 2 | Kitchen | MF-K | 4/3 | 128' | 20' | 169' | 2,093 | 0.42 gpm | 1.5 ft | 87°F |
| L11 | 2 | Kitchen | MF-K | 2/1 | 106' | 26' | 159' | 2,254 | 0.45 gpm | 1.6 ft | 87°F |
| L12 | 2 | Kitchen | MF-K | 10/9 | 156' | 15' | 187' | 607 | 0.20 gpm | 0.4 ft | 73°F |
| L13 | 2 | Kitchen | MF-K | 8/7 | 156' | 6' | 169' | 549 | 0.20 gpm | 0.4 ft | 73°F |
| L14 | 2 | Kitchen | MF-K | 6/5 | 156' | 12' | 181' | 587 | 0.20 gpm | 0.4 ft | 73°F |
| L15 | 2 | Kitchen | MF-5 | 2/1 | 144' | 9' | 164' | 777 | 0.20 gpm | 0.4 ft | 76°F |
| L16 | 2 | Pantry | MF-K | 12/11 | 107' | 26' | 161' | 1,550 | 0.31 gpm | 0.8 ft | 81°F |
| L17 | 2 | Entry, Office | MF-5 | 13/12 | 173' | 7' | 189' | 1,738 | 0.35 gpm | 1.2 ft | 79°F |
| L18 | 3 | Office | MF-5 | 16/17 | 122' | 17' | 156' | 1,400 | 0.28 gpm | 0.7 ft | 79°F |
| L19 | 2 | Living | MF-5 | 3/4 | 148' | 9' | 167' | 1,119 | 0.22 gpm | 0.5 ft | 79°F |
| L20 | 2 | Living | MF-5 | 5/6 | 148' | 13' | 174' | 1,171 | 0.23 gpm | 0.5 ft | 79°F |
| L21 | 2 | Living | MF-3 | 4/3 | 140' | 11' | 163' | 1,028 | 0.21 gpm | 0.4 ft | 76°F |
| L22 | 2 | Living | MF-3 | 1/2 | 152' | 3' | 159' | 1,015 | 0.20 gpm | 0.4 ft | 76°F |
| L23 | 2 | Living | MF-4 | 1/2 | 140' | 7' | 156' | 992 | 0.20 gpm | 0.4 ft | 76°F |
| L24 | 2 | Living | MF-4 | 11/12 | 152' | 3' | 159' | 1,015 | 0.20 gpm | 0.4 ft | 76°F |
| L25 | 3 | Bedroom 1 | MF-3 | 8/7 | 146' | 22' | 191' | 2,069 | 0.41 gpm | 1.6 ft | 80°F |
| L26 | 3 | Bedroom 1 | MF-3 | 6/5 | 146' | 15' | 178' | 1,844 | 0.37 gpm | 1.2 ft | 80°F |
| L27 | 3 | Closet (Bedroom 2), Bath 1 | MF-3 | 13/14 | 110' | 32' | 175' | 2,452 | 0.49 gpm | 2.0 ft | 87°F |
| L28 | 3 | Bedroom 2 | MF-3 | 18/17 | 126' | 12' | 152' | 1,124 | 0.22 gpm | 0.4 ft | 78°F |
| L29 | 3 | Bedroom 2 | MF-3 | 16/15 | 126' | 16' | 159' | 1,184 | 0.24 gpm | 0.5 ft | 78°F |
| L30 | 3 | Bath 2 | MF-4 | 3/4 | 136' | 8' | 153' | 908 | 0.20 gpm | 0.4 ft | 76°F |
| L31 | 3 | Bedroom 3 | MF-4 | 14/13 | 164' | 8' | 180' | 2,064 | 0.41 gpm | 1.5 ft | 81°F |
| L32 | 3 | Bedroom 3, Closet (Bedroom 3) | MF-4 | 10/9 | 128' | 16' | 160' | 2,381 | 0.48 gpm | 1.7 ft | 100°F |
| L33 | G | Garage | MF-G | 1/2 | 114' | 18' | 151' | 866 | 0.20 gpm | 0.4 ft | 56°F |
| L34 | G | Garage | MF-G | 3/4 | 112' | 17' | 147' | 842 | 0.20 gpm | 0.3 ft | 56°F |
| L35 | G | Garage | MF-G | 5/6 | 148' | 19' | 187' | 1,075 | 0.21 gpm | 0.5 ft | 56°F |
| L36 | G | Garage | MF-G | 7/8 | 150' | 16' | 184' | 1,057 | 0.21 gpm | 0.5 ft | 56°F |
| L37 | G | Garage | MF-G | 9/10 | 154' | 13' | 181' | 1,040 | 0.21 gpm | 0.5 ft | 56°F |
| L38 | G | Garage | MF-G | 11/12 | 154' | 11' | 176' | 1,010 | 0.20 gpm | 0.4 ft | 56°F |
| L39 | G | Garage | MF-G | 14/13 | 154' | 6' | 167' | 957 | 0.20 gpm | 0.4 ft | 56°F |

| Loop | Zone | Rooms | Manifold | Ports | In field | Leader | Total | Btu/h | Flow | Head | Floor |
|---|---|---|---|---|---|---|---|---|---|---|---|
| L40 | G | Garage | MF-G | 15/16 | 174' | 2' | 180' | 1,037 | 0.21 gpm | 0.5 ft | 56°F |
| L41 | G | Garage, Mechanical | MF-G | 32/31 | 154' | 9' | 172' | 1,244 | 0.25 gpm | 0.6 ft | 61°F |
| L42 | G | Garage | MF-G | 30/29 | 136' | 17' | 172' | 990 | 0.20 gpm | 0.4 ft | 56°F |
| L43 | G | Garage, Garage | MF-G | 27/26 | 106' | 29' | 165' | 948 | 0.20 gpm | 0.4 ft | 56°F |
| Zone 1 | 1 | Master suite | MF-1 | | 6 loops | | 880' | 7,622 | 1.5 gpm | | |
| Zone 2 | 2 | Great room and utility | MF-1, MF-2, MF-3, MF-4, MF-5, MF-K | | 17 loops | | 2,845' | 19,513 | 4.3 gpm | | |
| Zone 3 | 3 | Bedroom wing and office | MF-3, MF-4, MF-5 | | 9 loops | | 1,503' | 15,426 | 3.1 gpm | | |
| Zone G | G | Garage (held at 50°F) | MF-G | | 11 loops | | 1,882' | 11,066 | 2.3 gpm | | |

## Manifolds

Each station's row of 3" port positions along its wall; a loop takes two next to each other.

| Tag | Loops | Ports used | Where | Zones | Feed |
|---|---|---|---|---|---|
| MF-1 | 7 loops: L1, L2, L3, L4, L5, L6, L7 | 14 of 14 | Laundry, on the north wall west of the dryer | 1, 2 | 1" pair F-1, 41' each way |
| MF-2 | 2 loops: L8, L9 | 4 of 8 | Mud room, on the south wall (garage wall), west end | 2 | 1" pair F-2, 27' each way |
| MF-G | 11 loops: L33, L34, L35, L36, L37, L38, L39, L40, L41, L42, L43 | 22 of 34 | Garage, on the mud room wall (two 7-loop manifolds side by side) | G | 1" pair F-2, 27' each way (shared with MF-2) |
| MF-K | 6 loops: L10, L11, L12, L13, L14, L16 | 12 of 12 | Mud room, east wall (kitchen wall), south of the powder room | 2 | 1" pair F-K, 17' each way |
| MF-5 | 5 loops: L15, L17, L18, L19, L20 | 10 of 17 | Pantry, on the east wall (entry wall) | 2, 3 | 1" pair F-5, 12' each way |
| MF-3 | 7 loops: L21, L22, L25, L26, L27, L28, L29 | 14 of 18 | Bedroom hall, west wall north of the fireplace | 2, 3 | 1" pair F-3, 52' each way |
| MF-4 | 5 loops: L23, L24, L30, L31, L32 | 10 of 14 | Bedroom hall, west wall south of the fireplace | 2, 3 | 1" pair F-4, 43' each way |

## Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The installer takes off the final list.

| Item | Qty | Notes |
|---|---|---|
| 1/2" O2-barrier PEX, 1,000' coils | 8 coils | 7,109' in 43 uncut lengths of 118' to 191' (the schedule), cut longest first |
| EPS25 Super 4 nub board, 4' x 4' | 274 boards (55 bundles of 5) | 4,167 sq ft of slab + 5% for cuts; R-18, 1/2" tube channels on 3" centres |
| 1" PEX supply + return, insulated | 404 lf | HS-1 to the stations, in the base course under the boards |
| Stainless manifolds with flow meters | 7 stations | MF-1 7-loop, MF-2 2-loop, MF-G 11-loop, MF-K 6-loop, MF-5 5-loop, MF-3 7-loop, MF-4 5-loop (MF-G as two manifolds side by side) |
| 24 V thermal actuators | 43 | one per loop, on the return port |

| Item | Qty | Notes |
|---|---|---|
| Zone controller + thermostats | 1 + 4 | T1, T2, T3, TG; outdoor reset sensor |
| Circulators | 2 | P-R1 constant ΔP for the house, P-RG for the garage with its mixing valve |
| Manifold cabinets | 7 | recessed at MF-K, MF-5, MF-3 and MF-4; surface at MF-1, MF-2 and MF-G |
| R-10 rigid foam, 2", slab edge | 346 lf | the perimeter of the slab, full depth |
| 10-mil vapor barrier | 4,584 sf | under the boards, lapped and taped |
| Pipe insulation, 1/2" x 12" | 86 pcs | one sleeve on every tube end where it rises out of the concrete |